

Fazle Mawla Wahyuhanda

Artificial Intelligence Engineering Student

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PROFESSIONAL SUMMARY

Artificial Intelligence Engineering student with a strong foundation in machine learning, deep learning, and data analysis. Experienced in building practical AI solutions across computer vision, natural language processing, and data engineering. Disciplined, collaborative, and committed to delivering reliable, well-documented work.

EDUCATION

Institut Teknologi Sepuluh Nopember

Jun 2024 – Jun 2028 (Expected)

Bachelor of Artificial Intelligence Engineering, Surabaya

Grade Point Average: 3.76 / 4.00

PROJECTS

RESEARCH AND COMPETITION PROJECTS

Paddy Field Segmentation from Pléiades Satellite Imagery

DINOv3 ViT-L (Satellite), UNet++, UPerNet Swin-L, Mask2Former, PyTorch

Built a four-model segmentation ensemble for binary paddy field mapping on 1024×1024 Pléiades tiles with spatially disjoint test blocks, combining satellite-pretrained DINOv3, UNet++, UPerNet, and Mask2Former through probability averaging, flip test-time augmentation, and label-style morphological post-processing, reaching a validation IoU of 0.969 as a Hology 9.0 finalist.

Property Price Prediction from Listing Descriptions

DeBERTa-v3-large, ModernBERT-large, Hugging Face Transformers, Modal

Fine-tuned DeBERTa-v3-large and ModernBERT-large regressors to predict property prices from free-text listing descriptions across 14,640 training listings, combining multi-seed 5-fold runs through a log-space blend with frozen out-of-fold recalibration. Reduced error from a TF-IDF and Ridge baseline (OOF MAE 398,141) to a public leaderboard MAE of 313,285 in the Hology 9.0 preliminary round.

Cross-Model Residual Erasure (CMRE) for Source-Bias-Robust Unsupervised E-Waste Clustering

SigLIP2, DINOv3, PE-Core, PCA, Spectral Clustering, SCMax

Proposed Cross-Model Residual Erasure (CMRE) for label-free clustering of 3,792 e-waste images in the BDC Satria Data 2026 semifinal, projecting out the dominant directions of SigLIP2 and Perception Encoder embeddings that DINOv3 cannot predict to remove photo-source bias without object labels or style proxies. Combined with SCMax cluster-count selection and spectral clustering, CMRE raised cross-source consistency from 0.524 to 0.975 and reached AMI 0.917 on 2,329 labelled images, documented in a scientific paper.

Physis: Pediatric Wrist Fracture Triage

PyTorch, timm, FastAPI, Next.js, Supabase, Cloudflare R2

Developed an age-aware pediatric wrist fracture triage system that ranks radiographs for radiologist review using a PhysisViT classifier, calibrated probabilities, age-band percentiles, and a localisation-only Faster R-CNN detector. Built the end-to-end research and deployment pipeline across model training, inference API, and a clinical-facing web application.

Wikipedia Next-Click Prediction with OCR-Driven Graph Search

EasyOCR, OpenCV, LightGBM, NetworkX, BGE Embeddings

Built an OCR-driven hyperlink graph from 4,604 Wikipedia screenshot pages and combined policy-value ranking with leakage-aware, bounded graph search to predict a navigator's next click toward an unseen target article, achieving 0.7443 out-of-fold accuracy under target-disjoint GroupKFold validation.

Directed DCRNN-Style Traffic Speed Forecasting

PyTorch, GRU, Directed Diffusion, Adaptive Adjacency, Graph Neural Networks

Developed a graph-based residual forecasting model for 1,260 traffic sensors using outgoing and incoming diffusion, recurrent temporal encoding, and learned adaptive adjacency to predict traffic speed at 5, 10, and 15-minute horizons, reaching a best validation MSE of 30.43.

GO-JUDI: Weakly Supervised Web Mining for Online Gambling Promotion Triage

Web Mining, Weak Supervision, LLM Silver Audit, Heterogeneous Graph, PU Learning

Developed a decision-support pipeline over 21,192 crawled web pages from 972 domains using weak supervision, domain-disjoint LLM silver auditing, heterogeneous graph intelligence, and Positive-Unlabeled ranking. The link-feature model achieved F1 0.956, while PU ranking reached PR-AUC 0.955 for prioritizing hidden-risk candidates.

Explainable Clinical Actionability Mining of Cancer Genomic Variants

AACR Project GENIE, CIViC, LightGBM, XGBoost, SHAP

Built a pan-cancer actionability mining pipeline across 271,837 tumor samples and 3.32 million genomic alterations by integrating mutation, CNA, and fusion data with CIViC evidence to derive research-oriented rule-based labels. LightGBM achieved PR-AUC 0.7320 and recall 0.8368, with SHAP and permutation importance used for model interpretation.

Ambiguity-Aware Multi-Expert Waste Image Classification

ConvNeXtV2, DINOv3, SigLIP2, Patch-MIL, StratifiedGroupKFold

Designed a group-aware five-fold classification pipeline for Recyclable, Electronic, and Organic waste across 26,527 training images, combining global visual anchors with hierarchical and object-centric experts through a cross-fitted guarded router. A held-out pilot improved Macro-F1 from 0.987886 to 0.988332 without degrading the guarded evaluation subsets.

Cost-Aware LLM Routing for Open-Weight Qwen Models

Qwen, Modal, LLM-as-a-Judge, RouteLLM, TF-IDF, Ridge Regression

Built an LLM routing prototype that selects among three Qwen models served on Modal GPUs to balance answer quality against cost, benchmarked against cheapest-model and best-single-model baselines. Calibrated the router with blind LLM-as-a-judge scores on RouteLLM prompts under a train/test split and trained a TF-IDF and Ridge quality predictor for live routing.

Sitting Posture Screening with REBA Scoring

MediaPipe, RTMPose, Modal, Progressive Web App

Developed a progressive web app that screens sitting posture from a single side-view photo using REBA ergonomic scoring, comparing on-device MediaPipe pose estimation with RTMPose-L WholeBody served on a serverless GPU, while processing photos in memory without storing them.

Missing-Modality Ablation for 3D Brain Tumor Segmentation

PyTorch, BraTS 2023, 3D MRI, Modal

Designed a 3D brain tumor segmentation pipeline on BraTS 2023 MRI volumes that evaluates all 15 combinations of T1n, T1c, T2w, and FLAIR sequences to measure how missing modalities affect tumor sub-region segmentation, with cost-tracked training on cloud GPUs.

COURSE FINAL PROJECTS

Multimodal Deep Learning for Poverty Mapping

Deep Learning, Satellite Imagery, Natural Language Processing

Developed a multimodal deep learning model that estimates poverty levels at the sub-district level by combining satellite imagery features with poverty-related news indicators across Surabaya, Gresik, and Bangkalan. Built the spatial image branch and the temporal tabular branch, then fused them to produce yearly poverty risk estimates constrained by official regional statistics.

Retrieval-Augmented Generation for Poverty News and Documents Analysis

RAG, Hybrid Retrieval, Multilingual Embeddings

Designed a retrieval-augmented generation pipeline over an Indonesian poverty news corpus, covering document chunking, multilingual embedding, and hybrid retrieval to surface relevant context for sub-district narratives. Supported the poverty mapping project by turning unstructured documents and news into structured, traceable evidence.

Javanese Script OCR with Ensemble Learning

ConvNeXtV2, YOLOv8, Support Vector Machine

Built an end-to-end Javanese script recognition system that segments sentence images, classifies characters with an ensemble of models, and transliterates the result to Latin text using rule-based logic.

E-Commerce Data Pipeline and Analytics Dashboard

Airflow, PySpark, ClickHouse, Metabase, Docker

Built an automated end-to-end data pipeline that extracts e-commerce transactions through an API, processes them across data lake and data warehouse layers, and presents the results on an interactive analytics dashboard.

EXPERIENCE

Data Science Mentor, Information Systems Expo (ISE) 2026

Sep 2026

Department of Information Systems, Institut Teknologi Sepuluh Nopember, Surabaya

- Mentored participants in a multi-day data science bootcamp covering Python, exploratory data analysis, data preprocessing, and feature engineering.
- Taught machine learning for regression, classification, clustering, hyperparameter tuning with Optuna, and introductory deep learning through hands-on notebooks.
- Guided participants from problem framing to final project delivery, reviewing their modelling choices and evaluation methods.

Laboratory Administrator, Information Intelligent Management Laboratory (IIMLab / MCI Lab), ITS

Jun 2026 – Present

Institut Teknologi Sepuluh Nopember, Surabaya

- Managed laboratory administration and operational needs to support academic and research activities.
- Coordinated internal information, documentation, and communication related to laboratory members and ongoing agendas.
- Supported seminars, guest lectures, research activities, and internal programs through data management and administrative coordination.

Creative Design Director, Schematics 2026

Apr 2026 – Present

Surabaya

- Directed the creative design division and oversaw the visual identity for all events and campaigns.
- Managed a team of designers to deliver consistent assets across digital and print media.
- Established design guidelines to maintain brand consistency across all materials.

AI Research Freelance

Mar 2026 – Present

Independent Research Projects, Remote

- Developed and tested machine learning and deep learning pipelines across computer vision, natural language processing, and spatiotemporal forecasting tasks.
- Built applied AI prototypes in cost-aware LLM routing, pose-based ergonomic screening, and 3D medical image segmentation.
- Prepared, cleaned, annotated, and validated datasets, including structured data entry and quality checks for reliable experiments.
- Designed evaluation workflows using cross-validation, ablation testing, error analysis, and metric tracking to identify model weaknesses.
- Documented experiments and reproducibility steps, then iterated on preprocessing, model configuration, and inference workflows.

Secretary of Creative Media Information, HMTC 2026

Jan 2026 – Present

Surabaya

- Managed the end-to-end workflow of all bureaus within the department.
- Coordinated task delegation, scheduling, and progress tracking across multiple bureaus.
- Served as the main liaison between bureau heads and the executive board.

Secretarial Staff, Gerigi x UKM Expo ITS 2025

Jul 2025 – Aug 2025

Surabaya

- Managed participant data and documentation for accurate attendance records.
- Served as a mentor assistant, guiding new students throughout the event.
- Coordinated with mentors and divisions for smooth task distribution.

Mentor Staff, Inclenation 2025

Jun 2025 – Aug 2025

Surabaya

- Introduced the faculty and its culture to new students.
- Guided mentees throughout the event to keep them engaged and informed.
- Fostered a supportive atmosphere to help mentees adapt to the faculty.

Design Marketing Staff, Schematics 2025

Apr 2025 – Dec 2025

Surabaya

- Worked with the marketing team to plan and execute visual strategies for event promotion.
- Developed digital marketing materials to support event campaigns.
- Maintained the consistency of the visual identity across marketing channels.

Creative Design Staff, ELITE x PESRAF 2025

Apr 2025 – Oct 2025

Surabaya

- Created design assets to improve event visibility and participant engagement.
- Developed new design concepts with the creative team to support the branding vision.
- Collaborated to refine creative ideas for ongoing and upcoming projects.

Media Creative Staff, Heroes Day Esports ITS

Mar 2025 – Dec 2025

Surabaya

- Designed graphics and multimedia content for social media, promotions, and event branding.
- Collaborated on design concepts for a range of creative projects.
- Worked with other divisions to develop creative ideas and follow current design trends.

Data Management Staff, ATLAS x ILITS 2025

Nov 2024 – Feb 2025

Semarang

- Managed attendance records for committee members and participants.
- Organized and analyzed feedback to support event evaluation.
- Prepared official reports and documentation to maintain accountability.

Event Committee Member, Informatics Engineering 24 Committee

Oct 2024 – Dec 2025

Surabaya

- Organized event schedules to ensure smooth operations and participant engagement.
- Collaborated with team members to create memorable event experiences.
- Coordinated logistics and provided real-time problem solving during events.

Illustrator and Graphic Designer Intern, Paradox Studio

May 2023 – Dec 2023

Creative studio for 2D and 3D illustration, animation, and design, Semarang

- Produced digital illustrations aligned with client requirements.
- Collaborated on design concepts for a range of creative projects.
- Applied structured design methods to meet project deadlines.

ACHIEVEMENTS

2026 Finalist, Hology 9.0

2026 1st Honorable Mention, BDC Satria Data 2026

2026 1st Honorable Mention, Joints x Inspire UGM Data Science

2026 Delegate, GEMASTIK Data Mining 2026

2026 Semifinalist, UI Datathon 2026

CERTIFICATIONS AND TRAINING

2025 Leadership and Self-Management Training, Pre-Basic Level, LKMM TD V FTEIC

2024 Leadership and Self-Management Training, Pre-Basic Level, LKMM Pra-TD V FTEIC

2024 Scientific Writing Training, Basic Level, PKTI-TD

2024 Student Entrepreneurship Skills Training, Basic Level, LKMW-TD